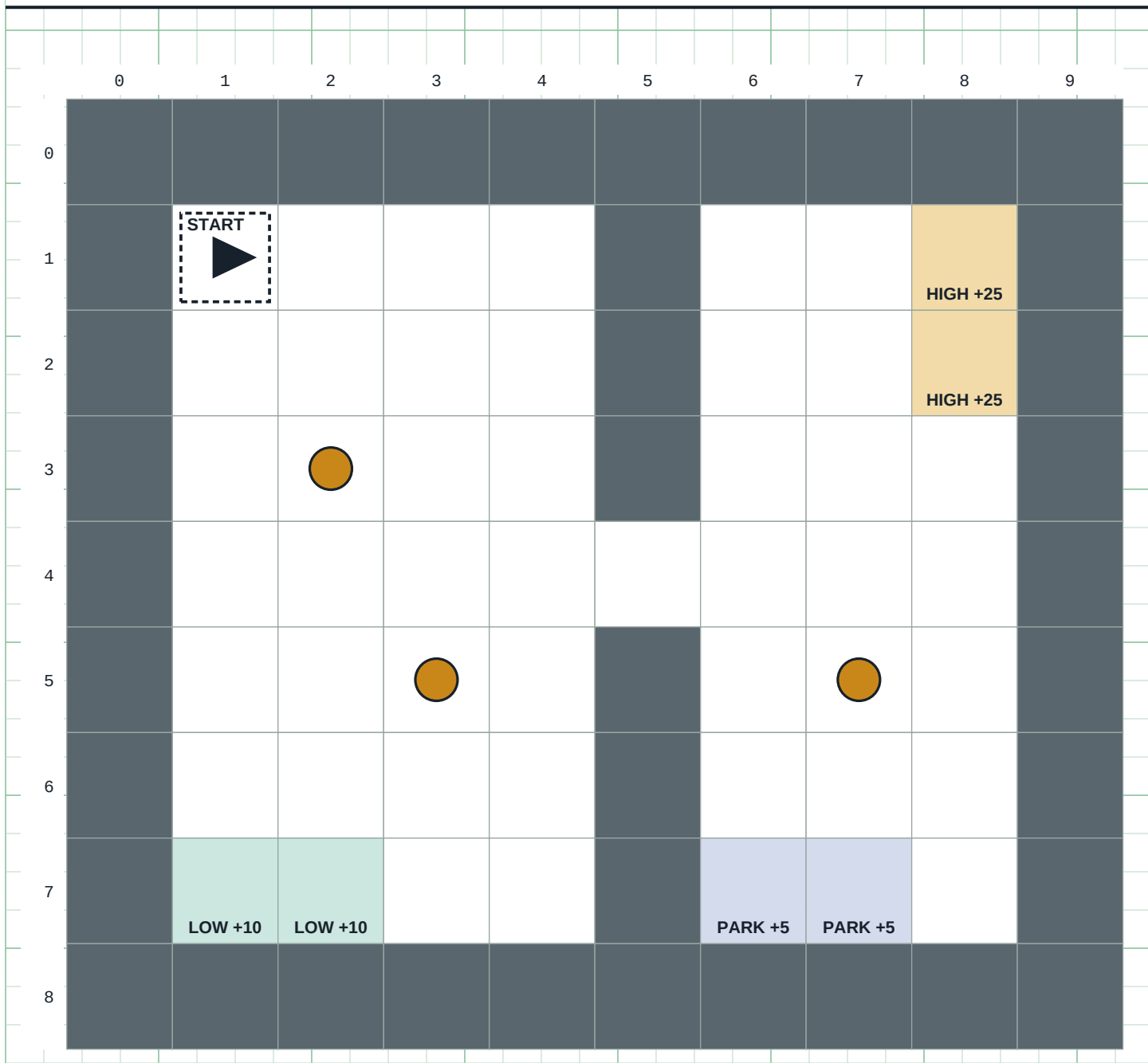


# FIELD SHEET

Study the field before you plan. Coordinates are (x, y). x counts across from the left, y counts down from the top.



Budget: 50 commands. Each command costs one, including turns and wasted moves.

Score: cargo on a LOW tile earns 10, on a HIGH tile 25. Parking on a PARK tile at the end earns 5.

The robot starts facing East. It carries one cargo at a time, and a tile holds one cargo.

Commands: forward, turnLeft, turnRight, grab, release.

TEAM

DATE

FIELD

SHEET

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# PLAN SHEET

Think first. Pencil only. Nothing here needs code.

## 1. GOAL: what is the most you can realistically score, and why?

## 2. OBJECTIVES: list everything on the field worth doing.

| Objective | Points | Est. commands | Order |
|-----------|--------|---------------|-------|
|           |        |               |       |
|           |        |               |       |
|           |        |               |       |
|           |        |               |       |
|           |        |               |       |
|           |        |               |       |
|           |        |               |       |

## 3. RISKS: what could waste commands or get the robot stuck?

## 4. APPROACH: say your plan in one or two sentences.

## 5. PREDICTION: write these down BEFORE you run anything.

Predicted points

Predicted commands used

Actual (fill in after)

TEAM

DATE

PLAN

SHEET

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# ROUTE SHEET

Write the robot's route one command at a time. Track where it ends up after each one.

| #  | Command | x | y | Facing |
|----|---------|---|---|--------|
| 1  |         |   |   |        |
| 2  |         |   |   |        |
| 3  |         |   |   |        |
| 4  |         |   |   |        |
| 5  |         |   |   |        |
| 6  |         |   |   |        |
| 7  |         |   |   |        |
| 8  |         |   |   |        |
| 9  |         |   |   |        |
| 10 |         |   |   |        |
| 11 |         |   |   |        |
| 12 |         |   |   |        |
| 13 |         |   |   |        |
| 14 |         |   |   |        |
| 15 |         |   |   |        |
| 16 |         |   |   |        |
| 17 |         |   |   |        |
| 18 |         |   |   |        |
| 19 |         |   |   |        |
| 20 |         |   |   |        |
| 21 |         |   |   |        |
| 22 |         |   |   |        |
| 23 |         |   |   |        |
| 24 |         |   |   |        |
| 25 |         |   |   |        |

| #  | Command | x | y | Facing |
|----|---------|---|---|--------|
| 26 |         |   |   |        |
| 27 |         |   |   |        |
| 28 |         |   |   |        |
| 29 |         |   |   |        |
| 30 |         |   |   |        |
| 31 |         |   |   |        |
| 32 |         |   |   |        |
| 33 |         |   |   |        |
| 34 |         |   |   |        |
| 35 |         |   |   |        |
| 36 |         |   |   |        |
| 37 |         |   |   |        |
| 38 |         |   |   |        |
| 39 |         |   |   |        |
| 40 |         |   |   |        |
| 41 |         |   |   |        |
| 42 |         |   |   |        |
| 43 |         |   |   |        |
| 44 |         |   |   |        |
| 45 |         |   |   |        |
| 46 |         |   |   |        |
| 47 |         |   |   |        |
| 48 |         |   |   |        |
| 49 |         |   |   |        |
| 50 |         |   |   |        |

TEAM

DATE

ROUTE

SHEET

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# BLANK ENGINEERING PAPER

Draw the field yourself, sketch routes, work out distances, or try a different idea.

|                       |                        |
|-----------------------|------------------------|
| TEAM                  |                        |
| DATE<br><b>SKETCH</b> | SHEET<br><b>4 of 4</b> |